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COMPUTER ARCHITECTURE WK9

1. Scenario Analysis: Memory Hierarchy in a Server Caching System

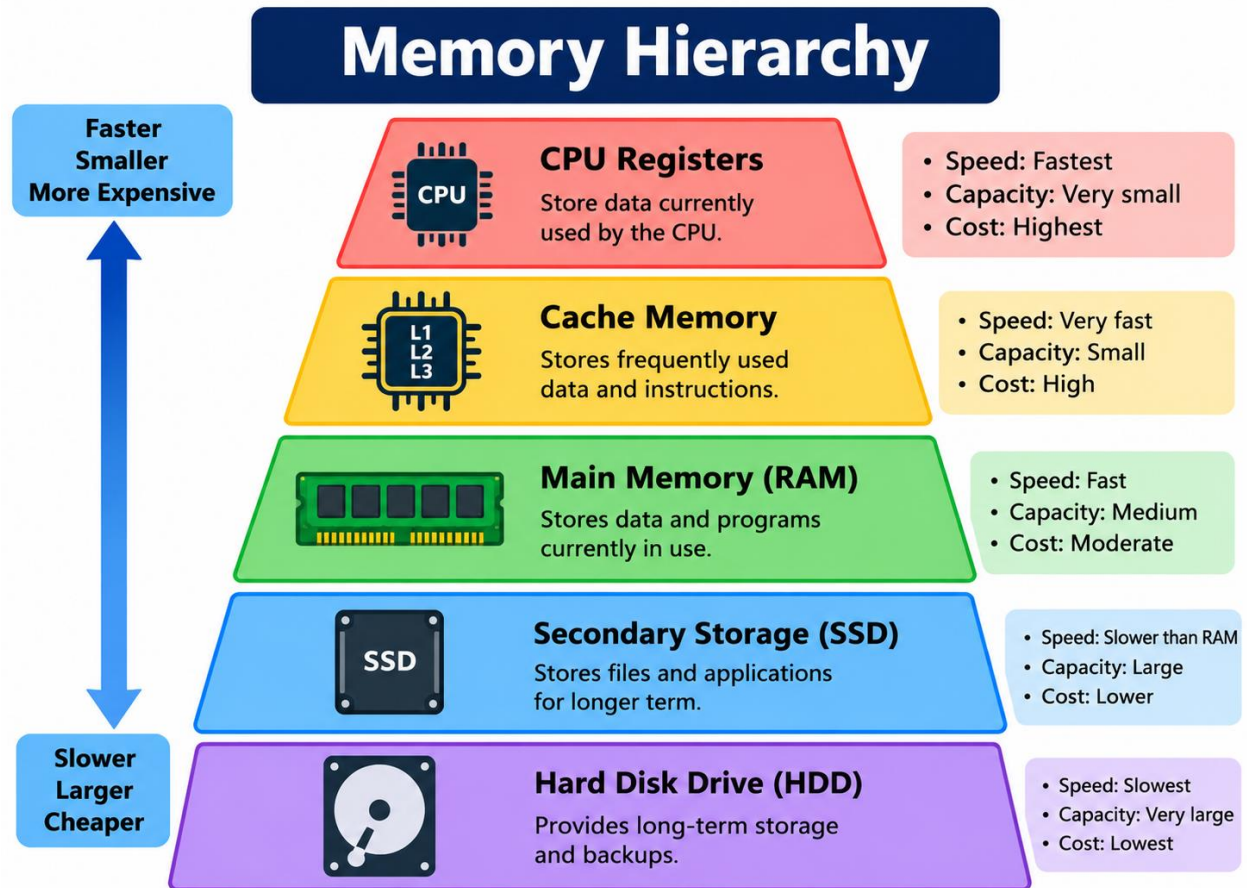
A memory hierarchy in a server caching system helps improve performance by storing frequently accessed data in different levels of memory. For example, an e-commerce website receives thousands of requests from customers searching for products. The server uses CPU cache to store frequently needed instructions and data for immediate access. RAM stores recently accessed product information, while an SSD stores larger amounts of data and files. A hard disk may be used for long-term storage and backups.

When a customer searches for a popular product, the server first checks the cache. If the information is available, it is returned quickly. If not, the server retrieves it from RAM or the SSD and may store it in the cache for future requests. This reduces access time, improves website responsiveness, and decreases the workload on storage devices. Therefore, an effective memory hierarchy helps the server handle more customers efficiently while providing a better user experience.

2. Concept Research: Cache Memory and Its Role

Cache memory is a small, high-speed memory located inside or close to the CPU. Its main role is to store frequently used data and instructions so that the processor can access them quickly. Cache memory reduces the time needed to retrieve information from the main memory (RAM), improving computer performance. It is commonly divided into three levels: L1, L2, and L3. L1 is the fastest and smallest, while L3 is generally larger but slower. Cache memory helps applications run smoothly, reduces CPU waiting time, and improves system efficiency. However, it is more expensive than RAM and has limited storage capacity.

3. TOOL PRACTICE:



I used Draw.io to create a memory hierarchy diagram showing the different levels of computer memory. The diagram includes CPU registers, cache memory, RAM, SSD, and hard disk storage. I arranged the components from fastest to slowest to demonstrate how the memory hierarchy works. Registers provide the fastest access, followed by cache memory and RAM. SSDs and hard disks provide larger storage capacities but have slower access speeds. This activity helped me understand the relationship between memory speed, cost, and capacity. Draw.io made it easy to organize the components clearly and present computer memory concepts visually for learning purposes.

4. Application Practice: Pseudocode for RAM Allocation

START

SET totalRAM = 1024

SET usedRAM = 0

```
INPUT requiredRAM
```

```
IF requiredRAM <= 0 THEN
```

```
    DISPLAY "Invalid memory request"
```

```
ELSE IF requiredRAM <= totalRAM - usedRAM THEN
```

```
    usedRAM = usedRAM + requiredRAM
```

```
    DISPLAY "Memory allocated successfully"
```

```
    DISPLAY "Remaining RAM: ", totalRAM - usedRAM
```

```
ELSE
```

```
    DISPLAY "Insufficient RAM available"
```

```
END IF
```

```
END
```

RAM allocation is the process of assigning available computer memory to programs and processes when they need it. The pseudocode demonstrates a simple memory allocation system using a total RAM capacity of 1024 MB. First, the system requests the amount of memory needed by a program. It checks whether the requested amount is valid and whether enough memory is available. If sufficient RAM exists, the system allocates the memory and displays the remaining capacity. Otherwise, it displays an error message. This process helps prevent memory over-allocation and improves resource management, allowing multiple applications to share computer memory efficiently.